

Relations

Chapter 9

Equivalence Relations

Section 9.5

Equivalence Relations

Definition 1: A relation on a set A is called an *equivalence relation* if it is reflexive, symmetric, and transitive.

Definition 2: Two elements a , and b that are related by an equivalence relation are called *equivalent*. The notation $a \sim b$ is often used to denote that a and b are equivalent elements with respect to a particular equivalence relation.

研究等价关系的目的在于将集合中的元素进行分类，选取每类的代表元素来降低问题的复杂度，如软件测试时，可利用等价类来选择测试用例。同班同学关系、同乡关系是等价关系。平面几何中三角形间的相似关系、全等关系都是等价关系。平面几何中直线间的平行关系是等价关系。

Strings

Example: Suppose that R is the relation on the set of strings of English letters such that aRb if and only if $l(a) = l(b)$, where $l(x)$ is the length of the string x . Is R an equivalence relation?

Solution: Show that all of the properties of an equivalence relation hold.

- *Reflexivity:* Because $l(a) = l(a)$, it follows that aRa for all strings a .
- *Symmetry:* Suppose that aRb . Since $l(a) = l(b)$, $l(b) = l(a)$ also holds, so bRa .
- *Transitivity:* Suppose that aRb and bRc . Since $l(a) = l(b)$, and $l(b) = l(c)$, then $l(a) = l(c)$ also holds and aRc .
- So R is an equivalence relation.

Congruence Modulo m

Example: Let m be an integer with $m > 1$. Show that the relation $R = \{(a,b) \mid a \equiv b \pmod{m}\}$ is an equivalence relation on the set of integers.

Solution: Recall that $a \equiv b \pmod{m}$ if and only if m divides $a - b$.

- *Reflexivity:* $a \equiv a \pmod{m}$ since $a - a = 0$ is divisible by m since $0 = 0 \cdot m$.
- *Symmetry:* Suppose that $a \equiv b \pmod{m}$. Then $a - b$ is divisible by m , and so $a - b = km$, where k is an integer. It follows that $b - a = (-k)m$, so $b \equiv a \pmod{m}$.
- *Transitivity:* Suppose that $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$. Then m divides both $a - b$ and $b - c$. Hence, there are integers k and l with $a - b = km$ and $b - c = lm$. We obtain by adding the equations: $a - c = (a - b) + (b - c) = km + lm = (k + l)m$. Therefore, $a \equiv c \pmod{m}$.

Divides

Example: Show that the “divides” relation on the set of positive integers is not an equivalence relation.

Solution: The properties of reflexivity, and transitivity do hold, but there relation is not transitive. Hence, “divides” is not an equivalence relation.

- *Reflexivity:* $a \mid a$ for all a .
- *Not Symmetric:* For example, $2 \mid 4$, but $4 \nmid 2$. Hence, the relation is not symmetric.
- *Transitivity:* Suppose that a divides b and b divides c . Then there are positive integers k and l such that $b = ak$ and $c = bl$. Hence, $c = a(kl)$, so a divides c . Therefore, the relation is transitive.

Equivalence Classes

Definition 3: Let R be an equivalence relation on a set A . The set of all elements that are related to an element a of A is called the *equivalence class of a* . The equivalence class of a with respect to R is denoted by $[a]_R$.

When only one relation is under consideration, we can write $[a]$, without the subscript R , for this equivalence class.

Note that $[a]_R = \{s \mid (a, s) \in R\}$.

If $b \in [a]_R$, then b is called a *representative of this equivalence class*. Any element of a class can be used as a representative of the class.

The equivalence classes of the relation congruence modulo m are called the *congruence classes modulo m* . The congruence class of an integer a modulo m is denoted by $[a]_m$.

$$[a]_m = \{ \dots, a - 2m, a - m, a, a + m, a + 2m, \dots \}.$$

For example,

$$\begin{aligned} [0]_4 &= \{ \dots, -8, -4, 0, 4, 8, \dots \} & [1]_4 &= \{ \dots, -7, -3, 1, 5, 9, \dots \} \\ [2]_4 &= \{ \dots, -6, -2, 2, 6, 10, \dots \} & [3]_4 &= \{ \dots, -5, -1, 3, 7, 11, \dots \} \end{aligned}$$

Equivalence Classes and Partitions

Theorem 1: let R be an **equivalence relation** on a set A . These statements for elements a and b of A are equivalent:

(i) aRb

(ii) $[a] = [b]$

(iii) $[a] \cap [b] \neq \emptyset$

Proof: We show that (i) implies (ii). Assume that aRb . Now suppose that $c \in [a]$. Then aRc . Because aRb and R is symmetric, bRa . Because R is transitive and bRa and aRc , it follows that bRc . Hence, $c \in [b]$. Therefore, $[a] \subseteq [b]$. A similar argument (omitted here) shows that $[b] \subseteq [a]$. Since $[a] \subseteq [b]$ and $[b] \subseteq [a]$, we have shown that $[a] = [b]$.

$$(i) \ aRb \qquad (ii) \ [a] = [b] \qquad (iii) \ [a] \cap [b] \neq \emptyset$$

Second, we will show that (ii) implies (iii). Assume that $[a] = [b]$. It follows that $[a] \cap [b] \neq \emptyset$ because $[a]$ is nonempty (because $a \in [a]$ because R is reflexive).

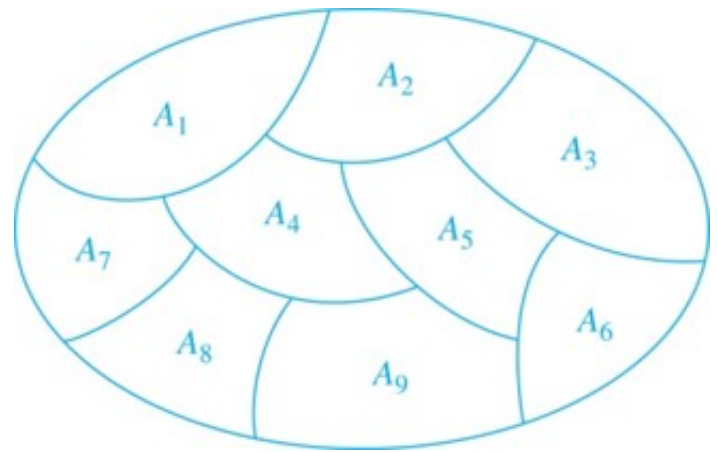
Next, we will show that (iii) implies (i). Suppose that $[a] \cap [b] \neq \emptyset$. Then there is an element c with $c \in [a]$ and $c \in [b]$. In other words, aRc and bRc . By the symmetric property, cRb . Then by transitivity, because aRc and cRb , we have aRb .

Because (i) implies (ii), (ii) implies (iii), and (iii) implies (i), the three statements, (i), (ii), and (iii), are equivalent. 

Partition of a Set

Definition: A *partition* of a set S is a collection of disjoint nonempty subsets of S that have S as their union. In other words, the collection of subsets A_i , where $i \in I$ (where I is an index set), forms a partition of S if and only if

- $A_i \neq \emptyset$ for $i \in I$,
- $A_i \cap A_j = \emptyset$ when $i \neq j$,
- and $\bigcup_{i \in I} A_i = S$.



A Partition of a Set

Partition of a Set

EXAMPLE 14

What are the sets in the partition of the integers arising from congruence modulo 4?

Solution: In Example 9 we found the four congruence classes, $[0]_4$, $[1]_4$, $[2]_4$, and $[3]_4$. They are the sets

$$[0]_4 = \{ \dots, -8, -4, 0, 4, 8, \dots \},$$

$$[1]_4 = \{ \dots, -7, -3, 1, 5, 9, \dots \},$$

$$[2]_4 = \{ \dots, -6, -2, 2, 6, 10, \dots \},$$

$$[3]_4 = \{ \dots, -5, -1, 3, 7, 11, \dots \}.$$

These congruence classes are disjoint, and every integer is in exactly one of them. In other words, as Theorem 2 says, these congruence classes form a partition. 

An Equivalence Relation Partitions a Set₁

Let R be an equivalence relation on a set A . The union of the equivalence classes of R is all of A , since an element a of A is in its own equivalence class $[a]_R$. In other words,

$$\bigcup_{a \in A} [a]_R = A.$$

From Theorem 1, it follows that these equivalence classes are either equal or disjoint, so $[a]_R \cap [b]_R = \emptyset$ when $[a]_R \neq [b]_R$.

Therefore, the equivalence classes form a partition of A , because they split A into disjoint subsets.

Section Summary

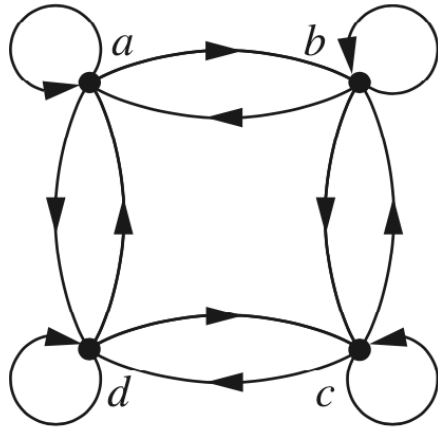
Equivalence Relations

Equivalence Classes

Equivalence Classes and Partitions

In Exercises 21–23 determine whether the relation with the directed graph shown is an equivalence relation.

23.



36. What is the congruence class $[4]_m$ when m is

a) 2?

41. Which of these collections of subsets are partitions of $\{1, 2, 3, 4, 5, 6\}$?

a) $\{1, 2\}, \{2, 3, 4\}, \{4, 5, 6\}$

b) $\{1\}, \{2, 3, 6\}, \{4\}, \{5\}$

46. Which of these are partitions of the set of real numbers?

d) the set of intervals $(k, k + 1), k = \dots, -2, -1, 0, 1, 2, \dots$

e) the set of intervals $(k, k + 1], k = \dots, -2, -1, 0, 1, 2, \dots$

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